Ecology

Perspective

FAIR and CARE in the age of agents: Design criteria for agent-ready environmental science

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Open Research Statement. Empirical data were not used for this research. Version-controlled manuscript source, citation-review metadata, and code used to render and audit the manuscript are available at https://github.com/CU-ESIIL/FAIR-and-CARE-for-AGENTS. A permanent archival identifier will be added before publication.

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Formatting proof. Editorial invitation, author declarations, finished figure, and an allowed Main Document format remain required.

Abstract

FAIR and CARE are not accommodations for AI. They are foundations for better science: research objects should be findable, accessible, interoperable, and reusable, while their use should advance collective benefit, respect authority, assign responsibility, and avoid harm. Human collaborators often approximate these commitments through a mixture of formal infrastructure and tacit knowledge. Coding and AI agents cannot be assumed to recover either. Their ability to generate plausible code or prose does not ensure that they will select authoritative inputs, preserve provenance, respect data governance, or stop when judgment and permission are required.

We argue that agentic workflows must therefore be designed to enact FAIR and CARE rather than merely instructed to "follow" them. We translate the principles into eight practical repository criteria and make them evaluable through **Goal → Instructions → Test → Record**. Each criterion pairs a design rule with repository evidence, a test, and an accountable human or community decision where automation is illegitimate. The objective is not to make repositories convenient for agents or to automate scientific judgment. It is to build workflows in which agents strengthen, rather than erode, the practices that make environmental science understandable, reproducible, accountable, and worthy of trust.

1. Better science requires designed-in FAIR and CARE

Environmental science is stronger when a project's purpose, evidence, methods, provenance, reuse conditions, responsibilities, and limits are explicit. Those qualities help collaborators assess claims, help reviewers trace evidence, help future researchers reproduce results, and help affected people understand or contest how data and scientific products are used. FAIR and CARE matter because they make these qualities part of scientific practice rather than optional documentation.

Environmental scientists increasingly ask coding agents to inspect repositories, edit analyses, debug workflows, prepare figures, or draft documentation. These systems can read files, execute code, call services, and change version-controlled artifacts. Yet most scientific repositories were built for people who already possess substantial project knowledge.

A familiar collaborator knows that *analysis_final_v2.R* is obsolete despite its name. They remember that a sensor threshold changed after an instrument failure, that one spreadsheet was edited manually, that the current figure comes from a different branch, or that culturally governed observations cannot be sent to an external model. This information may live in memory, messages, laboratory convention, or a conversation with the original analyst. It lets experienced people compensate for an underspecified repository.

An agent does not inherit this working knowledge, scientific judgment, or authority merely by gaining access to the repository. Give it a project URL with no previous conversation and ask: **Can it determine what the project is, how it works, what it should do, whether it succeeded, and what it is allowed to do?** Without a designed workflow, plausible action can mask the use of an obsolete file, an invalid method, an incomplete record, or an illegitimate data use.

The central design problem is therefore not how to make science easier for agents. It is how to prevent agentic work from bypassing the practices that make science reliable and legitimate. The workflow must carry scientific intent, authoritative inputs, methods, evaluation, provenance, permissions, and review gates forward into the agent's work. Building that infrastructure first improves science for people; it then gives agents an explicit scaffold within which they can contribute.

FAIR already makes machine actionability central to scientific data stewardship (Wilkinson et al., 2016). CARE complements object-centered stewardship with Collective Benefit, Authority to Control, Responsibility, and Ethics in the context of Indigenous Data Governance (Carroll et al., 2020). Agents do not replace these principles. They change how directly repositories can implement and test them. FAIR asks whether an independent actor can use the science correctly. CARE asks whether that actor should use it in that way. Test-first design asks how either claim will be evaluated.

Agent success does not prove that a scientific result is valid, and agent readiness does not imply that every task should be automated. Failed or refused tasks can still diagnose missing context, fragile infrastructure, absent evaluation, or functioning governance, but that diagnostic is a secondary benefit. The primary purpose is to design FAIR and CARE into the workflow before an agent acts.

2. Design the workflow before delegating the work

Telling an agent to be reproducible, responsible, or FAIR and CARE compliant is not an adequate control. The workflow itself must identify authoritative inputs, constrain permissible actions, define acceptable evidence, preserve provenance, and route consequential judgments to the people or communities with legitimate authority.

The operating rule is simple:

Define the task and the test before asking the agent to do the work.

We call this test-first scientific design. Before delegating consequential work, specify four elements:

1. **Goal -** What exactly should be accomplished, and what scientific claim or decision may the work support?
2. **Instructions** - Which data, methods, tools, degrees of freedom, and constraints apply? When must the agent stop or ask for review?
3. **Test -** What observable evidence distinguishes success, failure, and appropriate refusal?
4. **Record** - What inputs, code, environment, model, instructions, actions, outputs, evaluations, and approvals must be preserved?

The compact sequence is:

GOAL → INSTRUCTIONS → TEST → RECORD

A useful test has four dimensions. A **scientific** test asks whether the result addresses the question and meets domain expectations: for example, whether a model preserves mass balance or a reported confidence interval is reproduced. A **computational** test asks whether the environment builds, inputs validate, and the workflow completes. A **provenance** test asks whether another person can reconstruct what happened. A **governance** test asks whether the workflow was authorized to use those data, models, services, and publication channels.

Passing only three dimensions can still be failure. A scientifically correct result obtained through prohibited data use fails. A perfectly governed and reproducible hallucination fails. A correct result whose provenance cannot be reconstructed also fails this standard. Conversely, refusal can be the correct result of a governance test.

Consider an agent asked to flag suspect observations in a sensor network. The goal names the affected data product. The instructions identify the quality-control rules, authorized inputs, and operations the agent may propose but not publish. The test includes a labeled fixture, expected flag rates, and review of false positives. The record links the input version, code change, agent instructions, output, evaluation, and approving scientist. This specification is useful even if a person ultimately performs the analysis.

Tests need not all be software assertions. Some run automatically; others produce a review packet or block work pending an identified expert, data steward, or governance authority. What matters is that success and prohibited action are specified before an agent's plausible-looking output is available to influence the standard.

3. FAIR repository design for agents

FAIR describes properties of research objects and their stewardship, not a synonym for unrestricted openness. For an agent-ready repository, each principle can become a concrete design rule and a test.

F - Give every repository a front door

Design rule: Give every repository a front door. Use the pattern **one repository → one associated website**. The repository is the versioned scientific object; the website is its discovery and communication layer.

The landing page should state the scientific question and project scope before presenting a directory tree. It should identify the responsible people or organization, summarize data and methods, name major outputs, and link to the repository, manuscript or publications, preferred citation, and persistent identifiers. Descriptive titles, stable URLs, meaningful headings, ordinary search optimization, and machine-readable metadata help humans and tools orient themselves. The website should not duplicate every artifact. It should explain how distributed artifacts relate and which source is authoritative for each purpose. *(supporting source required before submission: machine-actionable environmental repository metadata)*

Test: Give a clean agent only the public project URL. Can it correctly identify the question, data, methods, outputs, repository, responsible people, version, and citation without unsupported guesses?

A - Give every agent an orientation

Design rule: Give every agent an orientation. Place an *AGENTS.md* at the repository root. It should explain what the project is, how the repository is organized, which workflows and outputs are canonical, how to run and test them, what constraints apply, what actions are prohibited, and when human approval is required.

Keep the file concise and link to detailed methods or governance documents rather than duplicating them. Preserve consequential prompts, decisions, or agent instructions when they materially affect methods, selection, interpretation, or dissemination. This is not an argument to log every chat: records may contain sensitive information and need deliberate scope, access, retention, and redaction. The governing rule is broader: **No scientifically necessary instruction should exist only in the memory of a person or an AI conversation.**

Test: Start a fresh agent with no previous conversation and ask it to perform a defined repository task. Record every additional fact a human must provide. Is that fact a documentation gap, a credential, a deliberate review gate, or a capability limitation?

I - Make scientific products portable

Design rule: Make scientific products portable. Agents should produce durable, editable artifacts rather than leave scientific products trapped in chat histories or proprietary interfaces. Appropriate formats may include Markdown, Python, R, YAML, JSON, CSV, Parquet, NetCDF, Zarr, GeoTIFF, and editable figure source. The choice should preserve scientific meaning through explicit schemas, units, coordinate systems, missing-value conventions, identifiers, and relationships.

Use a predictable directory structure that separates data references, reusable source, analyses, tests, prompts, results, documentation, and environments. Keep figures connected to editable source and source data; keep numerical results in machine-readable form, not only prose. FAIR principles for research software similarly emphasize identification, access to software and metadata, qualified references, and reuse conditions (Barker et al., 2022).

The model should be disposable; the science should persist.

Test: Move an artifact created with one agent or model to another agent and to a conventional computational environment. Can each understand, modify, execute, and reproduce it without proprietary conversion?

R - Make the project executable elsewhere

Design rule: Make the project executable elsewhere. A reusable project combines version control, tagged versions, licenses, tests, a documented reproduction command, and a container or reproducible environment specification with important dependencies pinned. Data and models do not all belong in Git. The repository should point unambiguously to the immutable or versioned inputs used, including identifiers, checksums, access procedures, licenses, and governance conditions.

Research compendia, dynamic documents, containers, and established reproducible-computing practices already connect narrative, code, inputs, environments, and repeatable results (Gentleman & Lang, 2007; Peng, 2011; Sandve et al., 2013; Boettiger, 2015). Agent-assisted work adds a need to identify consequential models or services and to preserve sufficient instructions and evaluation evidence. Interoperable provenance concepts such as entities, activities, agents, and their relationships can support that record (World Wide Web Consortium, 2013).

Test: Clone a tagged repository onto clean infrastructure, obtain the authorized input versions, and run one documented command. Can it reproduce a specified figure, table, statistic, or result within declared scientific tolerances?

4. CARE governance for agents

CARE originated in Indigenous Data Governance and cannot be reduced to a generic technical checklist. The criteria below describe ways repository and computational infrastructure can make established governance decisions operational. They do not create legitimate authority. Projects working with Indigenous data should follow the decisions and protocols of the relevant rights-holders and governance bodies, including when those decisions limit what becomes explicit or machine accessible. *(supporting source required before submission: nation- and community-specific governance protocols)*

C - State who benefits

Design rule: State who benefits. Before deploying an agentic workflow, identify its intended beneficiary, intended scientific or community benefit, expected useful output, and important burdens or risks. “Scientific progress” is too general when a workflow affects specific communities, contributors, field teams, or ecosystems.

A system that accelerates publication while increasing demands on data contributors may shift costs rather than create collective benefit. Benefit cannot be self-certified by a repository maintainer on behalf of a community. The record should identify who defined the benefit, how affected parties can evaluate it, and whether not automating is a legitimate outcome.

Test: Can the project state who should benefit, what observable outcome would constitute benefit, who will evaluate it, and what burdens must also be assessed?

A - Make authority explicit

Design rule: Make authority explicit. Data access is not blanket permission. Where relevant, distinguish authority to **read, copy, analyze, perform inference, train, fine-tune, combine, publish, and redistribute**. Associate governed data classes with allowed purposes, prohibited actions, required reviewers, retention limits, disclosure rules, and approved computational infrastructure.

Agentic workflows add a consequential question to data provenance. We routinely record where our data came from. Agentic science also requires knowing: **Where did the data go?** Governance requires knowing where computation occurs and which models and services receive the data. For sensitive ecological or community-governed data, record the compute location, identified model or service, model version when available, inference endpoint, provider retention and training behavior, network access, logs, and jurisdictional or institutional control. External transfer may require an approved enclave, institutional endpoint, community-approved compute, or local or self-hosted inference. Self-hosting does not itself establish legitimate use. It creates a computational boundary within which legitimate governance decisions can be enforced.

Technical policy should derive from legitimate governance and remain revisable by those with authority. Human approval should gate actions that cross data, institutional, publication, or community boundaries.

Test: Attempt a prohibited data movement or model use: for example, send restricted locations to an unapproved endpoint or use analysis-only data for training. Does the workflow prevent the action, explain the boundary, record the attempt, and escalate ambiguity to the correct authority?

R - Name the responsible human

Design rule: Name the responsible human. Every consequential agent workflow should have an identifiable human owner. The owner defines its authority, ensures that its tests remain relevant, validates important outputs, and decides whether evidence supports a scientific claim. Review is not meaningful if the reviewer cannot inspect the evidence or understand the evaluation.

For important outputs, preserve enough information to reconstruct the model or service, version when available, instructions, data, code, tools, environment, output, evaluation, and human review or authorization. Version control supports correction, but projects also need a route for reporting problems and withdrawing or amending consequential outputs.

Agents may receive autonomy, but responsibility cannot be delegated to them.

Test: Select an important result. Can we determine who authorized the workflow and reconstruct how the result was produced, tested, reviewed, and released?

E - Test what must not happen

Design rule: Test what must not happen. Do not leave ethics as a retrospective discussion. Before deployment ask: **What is the worst scientifically plausible thing this agent could do?**

For an environmental workflow, unacceptable outcomes might include disclosing a sensitive species location, fabricating literature support, publishing an unreviewed result, sending governed data to an unauthorized model, or presenting a fragile forecast as a high-consequence recommendation. Turn the most important cases into explicit prevention, detection, refusal, escalation, and recovery tests. Define affected parties and accountable owners, and include relevant scientific experts and rights-holders when designing and evaluating the cases.

Test: Deliberately attempt representative prohibited or harmful actions. Does the system refuse, detect, stop, or escalate appropriately-and can responsible people investigate and recover when a control fails?

5. An agent-ready environmental science repository

An ordinary laboratory can implement these criteria incrementally. The following architecture is illustrative, not mandatory:

```
my-environmental-project/
|-- README.md          # question, scope, people, outputs, citation
|-- AGENTS.md          # orientation, workflows, constraints, approvals
|-- environment/       # reproducible environment and dependency records
|-- data/
|   `-- README.md      # sources, versions, access, licenses, governance
|-- src/               # reusable analysis code
|-- analysis/          # declared workflows and editable analyses
|-- tests/
```

```
|   |-- scientific/        # domain expectations and result checks
|   |-- computational/     # setup, schema, smoke, and regression checks
|   `-- governance/        # prohibited actions and required escalation
|-- prompts/               # consequential task specifications or references
|-- provenance/            # run records: inputs, models, tools, reviews
|-- results/               # machine-readable outputs and editable figures
|-- manuscript/            # editable scientific narrative
`-- docs/                  # searchable associated website
```

The directories matter less than the functions they expose: scientific purpose, agent orientation, editable source, a reproducible environment, scientific and computational tests, governance boundaries, consequential instructions, provenance, outputs, and a public front door. A smaller project can begin with four changes:

1. Add a useful *README.md* and *AGENTS.md* that identify the project, canonical workflow, important result, and action boundaries.
2. Pin the important environment and data versions, then document one command that reproduces one result.
3. Add one scientific expectation and one governance boundary test.
4. Publish a landing page that connects the question, people, repository, data, outputs, and citation.

The first target should be bounded: "From a clean clone, reproduce Figure 1 using the declared environment and authorized test data; check the expected statistics; do not call unapproved network services; write a provenance record; and stop for review before changing a public artifact." A laboratory will learn more from making one workflow reliable and governable than from labeling an entire repository "AI ready."

6. Conclusion

FAIR and CARE are standards for better human science, not special accommodations for AI. They help make research understandable and reusable while keeping benefit, authority, responsibility, and ethics visible. Agentic systems create urgency because none of those commitments follows automatically from an agent's ability to read a repository, write code, or produce a convincing result.

The practical program is compact. Design **Goal → Instructions → Test → Record** into consequential work. Give the repository a front door and every participant an orientation. Require portable products and an executable project. State who benefits, make authority explicit, name the responsible human, and test what must not happen. Implement these practices first for one important result and one important boundary.

FAIR asks whether an independent actor can correctly use the science. CARE asks whether that use is beneficial, legitimate, and accountable. Test-first design makes both questions operational without pretending that code can replace scientific judgment or community authority. The aim is to build scientific workflows that help people do better science and ensure that agents participate within-rather than outside-the practices that make that science trustworthy.

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Author Contributions

Ty Tuff: Conceptualization, project direction, and writing - review and editing. This statement requires author confirmation before submission.

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References

Barker, M., Chue Hong, N. P., Katz, D. S., Lamprecht, A.-L., Martinez-Ortiz, C., Psomopoulos, F., Harrow, J., Castro, L. J., Gruenpeter, M., Martinez, P. A., & Honeyman, T. (2022). Introducing the FAIR Principles for research software. *Scientific Data, 9*, 622. https://doi.org/10.1038/s41597-022-01710-x

Boettiger, C. (2015). An introduction to Docker for reproducible research. *ACM SIGOPS Operating Systems Review, 49*(1), 71-79. https://doi.org/10.1145/2723872.2723882

Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., & Hudson, M. (2020). The CARE Principles for Indigenous Data Governance. *Data Science Journal, 19*, 43. https://doi.org/10.5334/dsj-2020-043

Gentleman, R., & Lang, D. T. (2007). Statistical analyses and reproducible research. *Journal of Computational and Graphical Statistics, 16*(1), 1-23. https://doi.org/10.1198/106186007X178663

Peng, R. D. (2011). Reproducible research in computational science. *Science, 334*(6060), 1226-1227. https://doi.org/10.1126/science.1213847

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLOS Computational Biology, 9*(10), e1003285.

https://doi.org/10.1371/journal.pcbi.1003285

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data, 3*, 160018. https://doi.org/10.1038/sdata.2016.18

World Wide Web Consortium. (2013). *PROV-O: The PROV Ontology. W3C Recommendation 30 April 2013*. https://www.w3.org/TR/prov-o/

Table 1. FAIR + CARE design and test matrix.

Principle	Design rule	What to implement	Test
F - Findable	Give every repository a front door.	One repository linked to one searchable website; question, people, data, methods, outputs, citation, and identifiers.	Give a clean agent only the project URL; score what it identifies correctly.
A - Accessible	Give every agent an orientation.	*README.md*, *AGENTS.md*, canonical workflows, commands, tests, constraints, prohibitions, and approval gates.	Start a fresh agent on a defined task; record every undocumented fact it needs.
I - Interope rable	Make scientific products portable.	Durable editable formats; schemas, units, identifiers, relationships, predictable structure, and source for outputs.	Move an artifact across agents and a conventional environment; modify and reproduce it.
R - Reusable	Make the project executable elsewhere.	Versioning, release, license, environment, pinned dependencies, input references, tests, provenance, and reproduction command.	Clone a tagged version on clean infrastructure and reproduce one named result.
C - Collective Benefit	State who benefits.	Beneficiary, purpose, expected output, benefit measure, burdens, evaluator, and option not to automate.	State the observable benefit and determine with affected parties whether it occurred.
A - Authority to Control	Make authority explicit.	Action-level permissions, approved compute and models, retention, network boundaries, review gates, and governance links.	Attempt prohibited movement or model use; verify prevention and escalation.
R - Respons ibility	Name the responsible human.	Workflow owner, run provenance, substantive review, authorization, disclosure, correction, and incident response.	Reconstruct an important result and identify who authorized and reviewed it.
E - Ethics	Test what must not happen.	Harm cases, affected parties, bias and disclosure checks, refusal rules, red-team tests, escalation, and recovery.	Attempt representative harmful actions; verify refusal, detection, response, and review.

Figure captions

Figure 1. A FAIR + CARE workflow makes scientific purpose, inputs, methods, evaluation, provenance, and authority inspectable. FAIR components make scientific objects usable and reusable; CARE components constrain whether a proposed use is beneficial, legitimate, and accountable. Test-first design connects both sets of principles to observable evidence and explicit human or community decisions. The same infrastructure first improves science for people and then gives agents an explicit scaffold for participating in it. The layout is illustrative rather than a required directory standard.